Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

a. Participant Name: Syed Naazim Hussain
b. Problem Statement: Build and deploy AI agents using Gemini, ADK, and Cloud Run.

Brief about the idea

Project Hermes: The Autonomous Content Factory

- **Hermes** is a production-grade multi-agent swarm that automates the entire lifecycle of YouTube Shorts — from trend discovery to published video with AI thumbnails and Twitter/X distribution.
- It transforms a high-level goal (e.g., "Create a video about sustainable tech") into a researched, A/B-tested, scripted, produced, and scheduled video, entirely without human intervention.
- **Multi-tenant by design:** Each user connects their own YouTube channel and Google Calendar via OAuth — all jobs, videos, and analytics are fully isolated per user.
- **Mission:** To bridge the gap between human creativity and autonomous execution using Agentic Workflows powered by Google ADK and Gemini.

Your solution should be able to explain the following:

- How did you approach the common problem statement and translate it into a working solution using ADK, MCP/AlloyDB AI while ensuring Hackathon's core requirements are met?
- What real-world problem does your solution address, and what practical impact does it create for users?
- What is the core approach or workflow behind your solution?

- **Approach:** We moved away from "Linear Pipelines" to "Collaborative Swarms." Using the **Google Agent Development Kit (ADK)**, we built a system where 7 specialized agents communicate via a shared state, allowing for recursive research, A/B script testing with Gemini evaluation, and self-correction during scriptwriting and production.
- **Real-World Problem:** Manual content creation is the #1 bottleneck for small creators and brands. Research, scripting, and editing take 6–8 hours per video.
- **Practical Impact:** Hermes reduces this to under 10 minutes of autonomous background processing, enabling creators to scale their voice without the burnout. Each run produces a YouTube Short, AI thumbnail, Twitter/X thread, and scheduling — all from one prompt.
- **Workflow:** Coordinator (Orchestrator) → Ideas (6-Source Discovery) → Research (Deep Web Synthesis) → ScriptWriter (A/B Testing + Twitter/X) → Producer (TTS + Gemini/Imagen/Flux.2 Visuals + Ken Burns + Whisper Captions + Thumbnails) → Scheduler (Calendar) → Analytics (Feedback Loop).

Opportunities

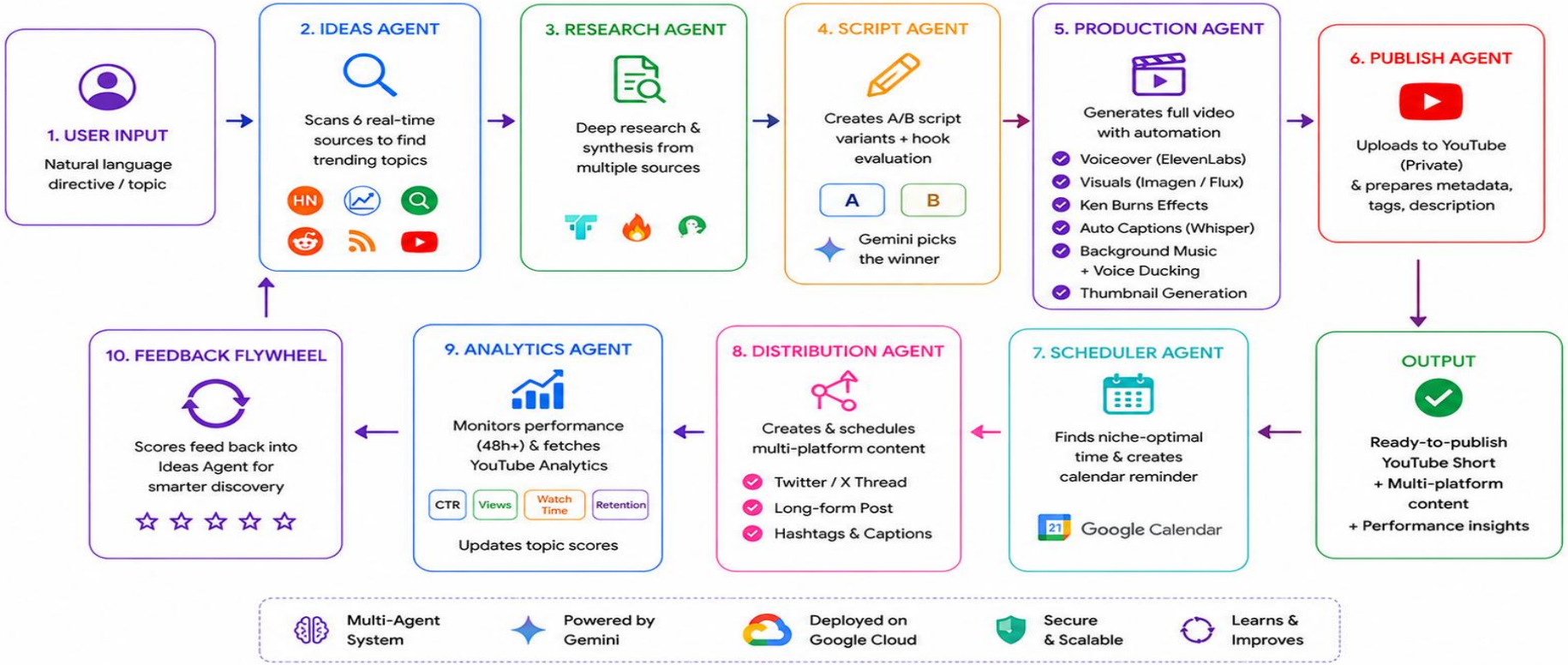
- How different is it from any of the other existing ideas?
- USP of the proposed solution

- Unlike existing "AI Editors" (which require human inputs at every step), Hermes is **fully autonomous**. It finds the trends for you using live data from 6 sources (HackerNews, Google Trends, DuckDuckGo, Reddit, RSS feeds, YouTube Trending), rather than just reacting to your text.
- **A/B Script Testing Built-In:** Hermes doesn't just write one script — it generates two competing variants (stat-bomb hook vs. curiosity-gap hook), has Gemini score them on 5 criteria (hook power, retention, niche fit, CTA strength, shareability), and automatically sends the winner to production.
- **Closed-Loop Analytics:** Most tools stop at "Publish." Hermes includes a 48h feedback loop that analyzes your channel's metrics (CTR, watch time, impressions, likes, comments) and automatically feeds those insights back into the "Ideas Agent" for next week's content. Topics that performed well get priority; 30-day deduplication prevents repeats.
- **Multi-Platform Output:** Every run produces a YouTube Short + AI thumbnail + Twitter/X thread (3-5 tweets) + long-form X post + hashtags — not just a video.
- **USP: Zero-Touch Video Production.** From Trend to YouTube Studio in one click — with per-user YouTube & Calendar OAuth, AI thumbnails, and niche-optimized everything.

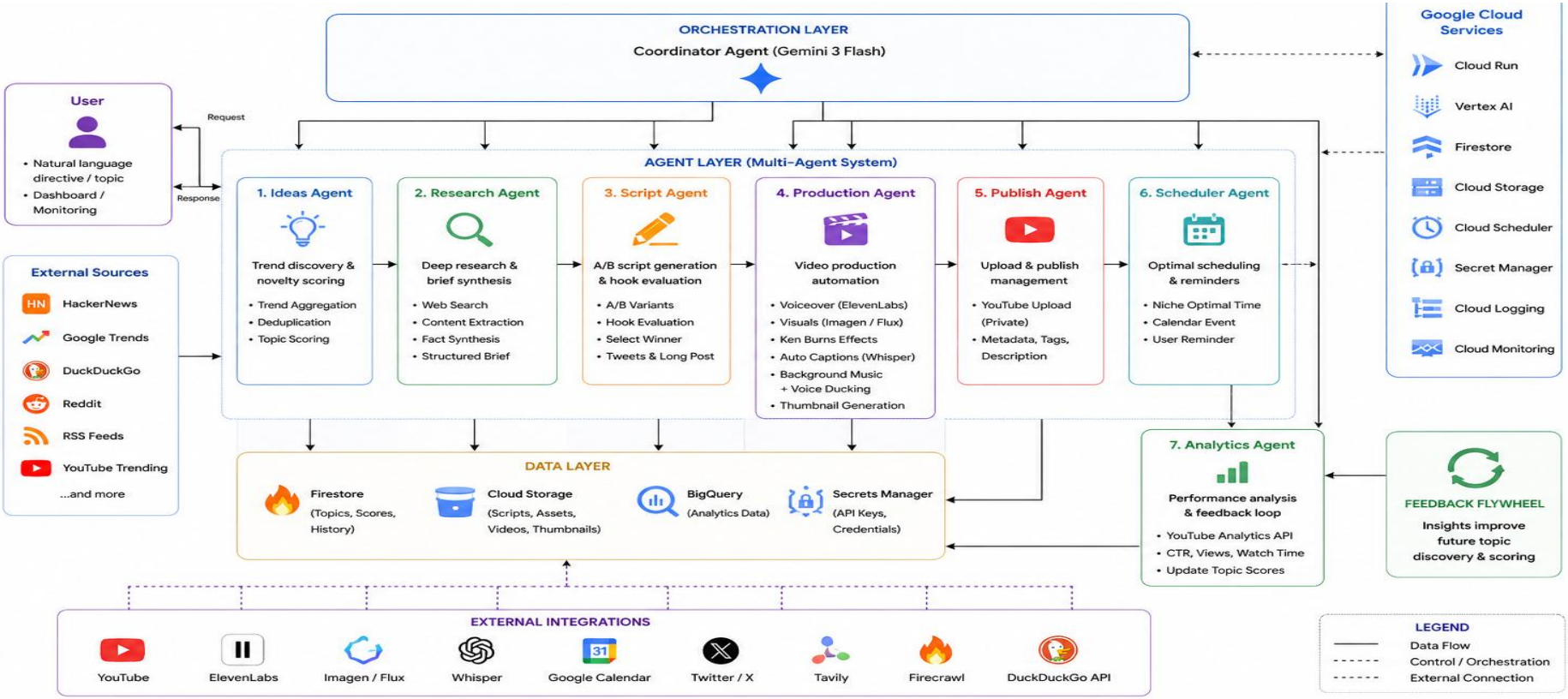
List of features offered by the solution

- **Agentic Orchestration:** 7 specialized agents with multi-agent task decomposition using Gemini 3 Flash via Google ADK 1.31+.
- **Live Trend Integration:** Real-time 6-source discovery via HackerNews, Google Trends, DuckDuckGo, Reddit, RSS feeds, and YouTube Trending — with 30-day novelty deduplication.
- **Recursive Research:** Fact-finding missions using Tavily, Firecrawl, and DuckDuckGo with structured brief output.
- **A/B Script Testing:** Two competing script variants auto-generated and evaluated by Gemini on 5 criteria — winner goes to production.
- **Multi-Provider Visuals:** Automated voiceovers (ElevenLabs / Edge TTS) and scene generation via Gemini 2.5 Flash Image, Imagen 3/4, or Flux.2 (Modal) — switchable via config.
- **Professional Video Assembly:** Ken Burns animation, Whisper word-level captions with niche-colored highlights, background music with voice ducking.
- **AI Thumbnail Generation:** Gemini/Flux.2-generated 16:9 thumbnails with Pillow bold text overlay and drop shadow.
- **Twitter/X Distribution:** Auto-generated thread (3-5 tweets) + long-form post + hashtags per video.
- **Smart Scheduling:** Per-user Google Calendar integration with niche-optimal posting windows.
- **Per-User Multi-Tenant Auth:** Firebase Auth (email/password + Google SSO) with per-user YouTube OAuth and Calendar OAuth — full data isolation.
- **Real-Time Dashboard:** SSE-powered live pipeline tracking, persistent chat with coordinator, creator profiles, retry upload, thumbnail preview.
- **Analytics Flywheel:** Automatic topic scoring based on YouTube performance — proven formats get priority in future runs.

Process flow diagram or Use-case diagram



Architecture diagram of the proposed solution:



Technologies/Google Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

The Core Stack

- **Gemini 3 Flash:** The "Central Brain." We chose Flash for its high-throughput and 1M+ context window, which is critical for ingesting hundreds of search results into a single coherent research brief. Also powers A/B hook evaluation (5-criteria scoring). Configurable — supports any OpenAI-compatible endpoint (e.g., Kimi-K2.5 via Nebius) via LiteLLM for flexibility.
- **Gemini 2.5 Flash Image:** Native image generation for scene visuals and AI thumbnails. Supports 9:16 portrait (Shorts) and 16:9 landscape (thumbnails) with retry and exponential backoff for rate limits.
- **Google ADK (Agent Development Kit):** Our orchestration engine. It allows us to build complex, tool-using agents that can self-correct and collaborate without hard-coded logic. 7 agents, each with typed Pydantic input schemas.
- **Google Cloud Run (Services + Jobs):**
- **Services:** Handle the API (40+ endpoints), Dashboard, SSE streaming, and real-time chat.
- **Jobs:** Specifically used for Media Production (ffmpeg/Imagen/Gemini Image/Whisper). This allows us to scale heavy processing tasks independently of the web server.
- **Cloud Firestore:** Our global "Shared Memory." It stores real-time agent state, per-user OAuth tokens (YouTube + Calendar), creator profiles, chat history, and analytics — allowing for persistent job monitoring even if a container restarts.
- **Google Cloud Storage:** All produced media (voiceovers, scene images, assembled videos, thumbnails) stored in GCS with signed URL generation via IAM service account impersonation.
- **Firebase Auth:** Multi-tenant user identity — email/password + Google SSO. Every job, video, token, and analytics record scoped to the authenticated user.

Technologies/Google Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

Why this Design?

- **Scalability:** By moving the "Production Agent" to a Cloud Run Job, we can generate 1,000 videos simultaneously. Each job gets its own dedicated 2-CPU/2GB environment, ensuring zero performance interference.
- **Economic Efficiency:** A purely serverless architecture means Zero cost when idle. We only pay for the exact seconds the agents are "working."
- **Multi-Provider Resilience:** Image generation supports 3 providers (Gemini native, Imagen, Flux.2) — if one hits rate limits or quota, the system can switch. TTS has ElevenLabs → Edge TTS fallback. Research has Tavily → Firecrawl → DuckDuckGo fallback chain.
- **Real-World Reliability:**
- **Fault Tolerance:** If a research agent fails due to an external API error, the ADK Runner can automatically retry the tool-call. Image generation includes exponential backoff (10s → 20s → 40s) for 429/RESOURCE_EXHAUSTED errors.
- **Graceful Degradation:** Database falls back to in-memory store when Firestore is unavailable. Storage falls back to local paths when GCS is unavailable. TTS falls back from ElevenLabs to Edge TTS.
- **Per-User OAuth Isolation:** Each user's YouTube and Calendar tokens are stored separately in Firestore — no shared global credentials. The production agent resolves credentials in order: per-user token → global fallback → clear error with instructions.
- **Global Deployment:** Containerization allows us to deploy Hermes near the creator, reducing latency for uploading video files to YouTube. Vertex AI Express keys supported for higher quotas in production.

Snapshots of the prototype

Job Tracker

API Docs | ✦ New Pipeline

All Jobs ↺ Refresh

- Create a video on how India achieved inde... ✓ done Apr 29, 08:31 AM
- Create a video about how electric cars work ✓ done Apr 29, 08:16 AM
- Create a video explaining how Machine lear... ✓ done Apr 29, 07:44 AM
- Create a video explaining how LLM work ✓ done Apr 29, 07:22 AM
- Valve's Surprise Comeback: Everything We... ✓ done Apr 29, 06:34 AM
- Why AI Isn't Replacing You Yet: Nvidia's Re... ✓ done Apr 29, 05:44 AM
- Create a Youtube video about ChatGPT 5.5... ✓ done Apr 29, 12:48 AM
- Create a Youtube video about ChatGPT 5.5...

✓ done Job bf217e2d-872… Apr 29, 08:16 AM

REQUEST
Create a video about how electric cars work

Pipeline | Script | 𝕏 Twitter

Ideas Discovery ✓
Trending topic selected

Deep Research ✓
Sources researched, facts extracted

Script Generation ✓
60-75s hook-first script written

Video Production ✓
TTS → Images → ffmpeg → YouTube

Coordinator Output

The YouTube Short content pipeline for **"How Electric Cars Work"** has been successfully completed.

✅ **Step 1: Idea & Validation**

- **Topic:** How Electric Cars Work
- **Rationale:** Topic validated and prioritized for the education niche.
- **Topic ID:** `2f6c3412-a0d5-4d87-987f-6b807a1127b7`

📌 **Step 2: Research**

- **Summary:** Explored the efficiency gap between ICE and EVs, the simplicity of electric motors (20 vs. 2,000 parts), and the role of the inverter and regenerative braking.
- **Key Fact:** EVs are ~77% efficient, while gas cars are only ~20% efficient.
- **Brief ID:** `d81458f8-80bf-4f31-856a-a145e3d3b6d6`

📝 **Step 3: Scripting & Social Content**

- **Winner Hook:** "Did you know a gas car wastes 80% of its fuel just to stay running?"

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Thank you